

- 10 (a) Explain why, for the photoelectric effect, the existence of a threshold frequency provides evidence for the particulate nature of electromagnetic radiation, as opposed to a wave theory.

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..... [4]

- (b) An electron is accelerated from rest in a vacuum through a potential difference of 4.7 kV.

- (i) Calculate the de Broglie wavelength of the accelerated electron.

wavelength = m [4]

- (ii) By reference to your answer in (b)(i) suggest why such electrons may assist with an understanding of the crystal structure.

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(c) The emission spectrum of isolated hydrogen atoms consists of a number of discrete wavelengths.

(i) Explain how the spectrum of hydrogen provides evidence for the existence of discrete energy levels in atoms.

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(ii) Some electron energy levels of the hydrogen atom are illustrated in Fig. 10.1.

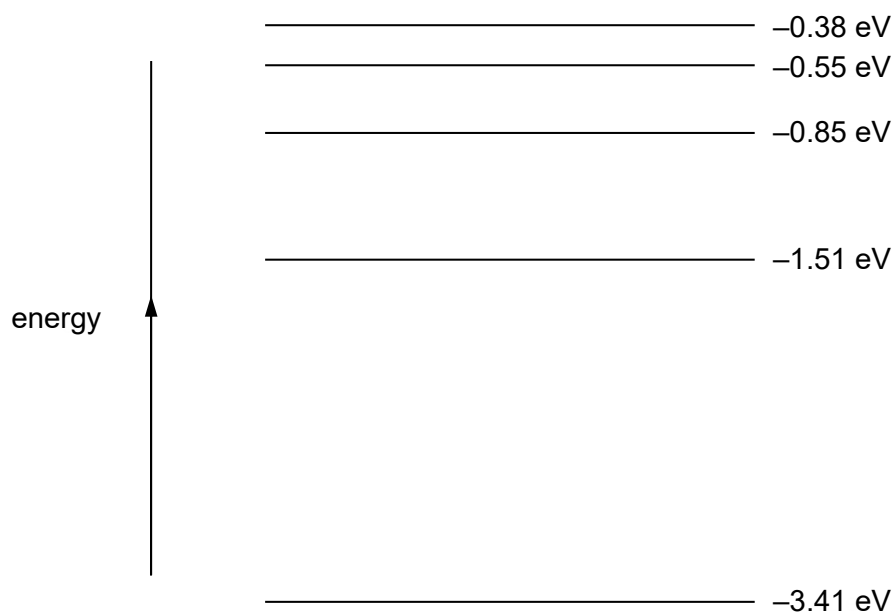


Fig. 10.1 (not to scale)

The minimum energy of a photon of visible light is 1.78 eV.

1. On Fig. 10.1, draw the transition giving rise to the longest wavelength in the visible spectrum. [1]
2. Use Fig. 10.1 to determine the shortest wavelength of the photons emitted.

wavelength =m [2]

- (iii) The light in a beam has a continuous spectrum of wavelengths from 400 nm to 700 nm. The light is incident on some cool hydrogen gas, as illustrated in Fig. 10.2.

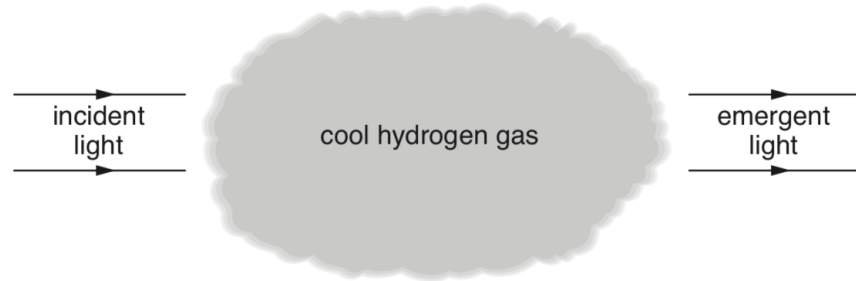


Fig. 10.2

Using the energy levels of the hydrogen atom in Fig. 10.1, state and explain the appearance of the spectrum of the emergent light.

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